

SOCluster - Towards Answering Unanswered Questions on Stack Overflow via Answered Questions

Abhishek Kumar, Deep Ghadiyali, Sridhar Chimalakonda and Akhila Sri Manasa Venigalla

Research in Intelligent Software & Human Analytics (RISHA) Lab

Department of Computer Science & Engineering

Indian Institute of Technology Tirupati, India

{cs17b002,cs17b011,ch,cs19d504}@iittp.ac.in

ABSTRACT

Stack Overflow (SO) platform has a huge dataset of questions and answers driven by interactions between users. But the count of unanswered questions is continuously rising, which is observed in various similar community Question & Answering platforms (Q&A) such as *Yahoo*, *Quora* and so on. To address this issue, these communities have explored clustering mechanisms to answer unanswered questions using other answered questions in the same cluster, which could also improve the response time for new questions. It is here, we propose *SOCluster*, an approach and a tool to cluster SO questions using a graph-based clustering approach. We selected four datasets of 10k, 20k, 30k & 40k SO questions without code-snippets or images involved, and performed clustering on them. We have done a preliminary evaluation of our tool by analyzing the resultant clusters using the commonly used metrics of *Silhouette coefficient*, *Calinski-Harabasz Index*, & *Davies-Bouldin Index*. We performed clustering for 8 different threshold similarity values and analyzed the intriguing trends reflected by the output clusters through the three evaluation metrics. At 90% threshold similarity, it shows the best improvement for the three evaluation metrics on all four datasets. We further manually assessed the content in the clusters to confirm the similarity of elements in clusters. This revealed clusters to correspond to topics such as *mouse over effect*, *speed optimisation*, *how to perform 'some' action in JavaScript*, and so on. The source code and tool are available for download on Github at: <https://github.com/rishalab/SOCluster>, and the demo can be found here: https://youtu.be/Ewm-M_rg_x8.

CCS CONCEPTS

• **Software and its engineering** → **Software notations and tools**.

KEYWORDS

stack overflow, question answering, unanswered, clustering, intent

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1 INTRODUCTION

Stack Overflow (SO) is one of the most commonly used Stack Exchange network focused on questions related to programming [8, 24]. SO's active community attracts information seekers from around the globe harvesting its knowledge base[1]. Despite the rapid growth, there has been a huge rise in the number of unanswered questions on SO [1], currently standing at more than 7.1 million¹, out of the total 23 million² questions, as of January, 2023.

Existing studies in literature [1, 15] identify that questions which are *too short*, *hard to follow*, *duplicate questions* and which *fail to attract an expert member* account for more than 50% of unanswered questions. Due to the fast-growing user base of SO (currently at around 14 million), the number of questions on the platform are rising, causing an increased load on the expert members, resulting in many questions not getting required attention. To maintain the popularity of the platform and to enhance responsiveness of such services, similar questions could be identified and relevant answers from the existing knowledge base of SO platform could be returned [11]. One of the most beneficial solutions to manage big amount of data is to cluster them automatically according to similarities [5, 24]. Clustering of content on SO was explored to identify relevant users with suitable domain knowledge towards answering unanswered questions [8]. The clustered questions can be leveraged to answer unanswered questions using other answered questions in that cluster.

Our goal is to cluster questions on SO platform based on *intent* as a way to help programmers. The below example explains why focusing on *intent* is important in the context of SO platform.

- *Not getting output for merge sort*³
- *A bug in merge sort*⁴

Both the above questions correspond to users asking about some error/bug present in their implementation of *Merge Sort*. Although the questions look different from each other at the outset, they have similar *intent* and so, the answer of one question might be used to answer the other question if they are in the same cluster.

¹<https://stackoverflow.com/questions?tab=Unanswered>

²<https://data.stackexchange.com/>

³<https://bit.ly/3IUU1ew>

⁴<https://bit.ly/3pOxJgX>

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The idea of clustering of web queries based on *intent* has been extensively explored in the literature [7, 12, 19, 21]. However, web queries can be a collection of random keywords, whereas questions asked to a community for help generally have semantic meaning. For example, “*python lambda function*” can be a web query but to ask on a Q&A site, one needs to write “*What is Lambda function in Python and how to use it?*”. But the distinction of web queries from community Q&A requires a different approach (which takes into consideration the semantic meaning) for clustering of questions on a community platform dataset such as the SO.

There have been multiple attempts towards clustering of questions on community platform datasets. Haponchik et al. [6] have clustered the *Quora* corpus using LSSVM, by training the model on pairwise annotated quora data. Chen et al. [3] used co-training approach to cluster *Yahoo! Answers*. Yelmen & Duru [24] clustered SO questions using *doc2vec* vectorizer, however recent BERT models have outperformed *doc2vec* on NLP tasks [13].

While there is existing work on clustering SO questions [8], they do not focus on identifying relevant answers to unanswered questions on SO, which is the core idea of our approach and tool. There have been various recent works focusing on tasks such as intent-recognition, intent-classification and so on using the latest BERT models [4, 9]. Huggins et al. confirm that BERT models are quite good in above tasks and achieve 94% accuracy with minimal training examples [9]. In this work, we leverage Sentence-BERT, a modification of the pretrained BERT network that uses siamese and triplet network structures to derive semantically meaningful sentence embeddings [17]. To the best of our knowledge, we are not aware of any approaches that use Sentence-BERT on SO to answer unanswered questions.

Driven by the recent advances in clustering methods based on graphical representations of the relationships among data points [14] and BERT models in intent recognition tasks, we propose

(i) a graph-based algorithm for intent-based clustering of SO questions and (ii) SOCluster, a tool based on Sentence-BERT vectorizer for creating intent-clusters.

2 SOCLUSTER - DESIGN AND DEVELOPMENT

SOCluster can be divided into three main steps as shown in Figure. 1. Firstly, we downloaded the SO data dump and processed it using a MySQL script to create well-organized SQL tables. We then filtered and vectorized the dataset using *Sentence-BERT*, which uses recent advances in NLP & NLU to generate sentence-embeddings [17]. In the next step, we create a weighted undirected graph where the questions are represented as nodes and edge-weight is calculated using the cosine similarity between vector representation of two questions. Lastly, the Clustering algorithm (explained in Algorithm 1) breaks this graph into multiple components by considering only those edges as valid whose weight is greater than the provided threshold similarity. Each of the resulting component in this undirected graph refers to a *cluster*.

2.1 Step 1: Dataset Generation and Pre-processing

2.1.1 Data dump. We downloaded SO post data and schema from StackExchange data dump archives⁵ [on Sept 2020], and organized them as tables using custom scripts. We then organised these files in MySQL database in the form of tables using custom scripts.

2.1.2 Pre-processing. We filtered the database by omitting incorrect or inconsistent data. We ignored questions containing *image*, *table*, *large code snippet* (multiple functions/methods), and so on, to limit the scope of our work, owing to processing overhead involved in image processing and in parsing the code snippets. Hence, we focused only on questions with no code snippet involved. In our dataset, the top two tags with most number of questions were *JavaScript* and *Python* with 2,081,682 and 1,528,649 questions respectively. Hence, we extracted a subset of the database on *JavaScript* and *Python* tags to perform our experiment, which included both answered as well as unanswered questions. We varied this subset size from 10000 to 40000 questions in four stages.

2.1.3 Feature Vectorization. SOCluster uses *Sentence-BERT* for feature vectorization as it considerably improves on BERT model [17], which in itself has performed significantly better compared to *GloVe* and *InferSent* in intent-classification tasks [18]. Attention, Transformers, BERT & Siamese Network are the four key concepts of Sentence-BERT [17]. The pooling layer in its architecture generates the embedding. It reduced the effort for finding the most similar pair in a collection of 10,000 sentences from 65 hours with BERT/roBERTa to about 5 seconds, while maintaining the accuracy from BERT [17]. The number of hidden units in default BERT model architecture is 768 and hence, the output of this vectorization stage is a 768-dimensional vector for each question. BERT models perform good in intent recognition and intent classification tasks as shown in literature [4, 9]. Therefore, we use BERT-embeddings to capture the intent of the questions and generate feature vectors.

2.2 Step 2: Graph Construction

2.2.1 Similarity Index. We used Cosine similarity in our tool to calculate the similarity between two vectors $\vec{A}$ and $\vec{B}$. It is measured by the cosine of the angle(ϕ) between two vectors. For our work, we chose Cosine similarity as the similarity index over *Euclidean distance*, *Jaccard coefficient*, *Pearson correlation coefficient* and other metrics owing to its popularity and usage in clustering and recent NLP approaches [10] [16]. Many recent works have used cosine similarity in the field of text document clustering such as Jalal et al. [10] to cluster text documents and Rao et al. [16] to design an end-to-end term/document clustering pipeline.

2.2.2 Graph generation. We created a weighted undirected graph using the feature vectors obtained as nodes and cosine similarity between them as the edge weights. We have used an adjacency matrix representation to store the graph.

2.3 Clustering

This step explains the Graph-based Clustering algorithm that SOCluster uses (Algorithm 1). It takes the graph generated in the last step

⁵<https://archive.org/download/stackexchange>

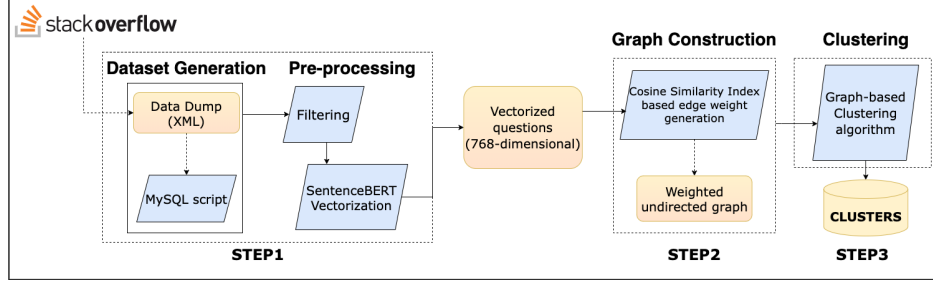


Figure 1: Architecture Diagram of SOCluster

as one of the inputs, along with a *Threshold Similarity Value*. It considers only those edges as valid, whose weight is greater than the threshold value, thereby breaking the graph into multiple components. On finding any unvisited node, it uses *Breadth-first Search (BFS)* traversal to search for the connected component and marks the already traversed nodes as *visited*, while edges with lesser weight than the threshold are ignored. Each component returned by *BFS* procedure is a *cluster*, and the algorithm returns a set of clusters as output.

Algorithm 1: Clustering algorithm

Input: A weighted undirected graph $G(V, E, W)$, Threshold similarity T_s

output: Set of Clusters

Let $Cluster_Set$ be a Set
 $Cluster_Set = \{\}$;
for each vertex $k \in V$ **do**
 $Visited[k] = False$;
end
for each vertex $k \in V$ **do**
 if $Visited[k] == False$ **then**
 $Cluster = BFS(V, k)$;
 $Cluster_Set = Cluster_Set \cup Cluster$;
 end
end
print $Cluster_Set$;
Procedure $BFS(V, k)$
 $Visited[k] = True$;
 Let Q be a Queue
 $Q.insert(k)$;
 Let $Cluster'$ be a Set
 $Cluster' = \{k\}$;
 while Q is not empty **do**
 $s = Q.front()$;
 $Q.pop()$;
 for each vertex $p \in V.Adj[s]$ **do**
 if $Visited[p] == False \ \&\& \ W[s][p] \geq T_s$ **then**
 $Visited[p] = True$;
 $Q.insert(p)$;
 $Cluster' = Cluster' \cup p$;
 end
 end
 end
end
return $Cluster'$;

3 PRELIMINARY EVALUATION

To evaluate SOCluster’s performance, we clustered 4 datasets of size 10k, 20k, 30k & 40k. Since, the clustering algorithm used in SOCluster takes *Threshold Similarity* as one of the inputs (as presented in Algorithm 1), we performed the experiment by changing this parameter over eight different values : 0.5, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85 & 0.9, for each dataset and observed the results.

To evaluate these outputs, we used three commonly used clustering evaluation metrics in the literature, which are used for clustering

performance evaluation when ground truth labels are unavailable [20]. The *Silhouette Coefficient* estimates the similarity of an object to its cluster compared to the next-nearest cluster, *Calinski-Harabasz Index* score is defined as the ratio of the sum of between-clusters dispersion and inter-cluster dispersion for all clusters and the *Davies-Bouldin Index* evaluates the cluster using quantities and features inherent to the dataset. We calculated these three metrics for clusters obtained for varied dataset sizes and threshold values, which helped us in narrowing down on the threshold values.

We also manually analysed the clusters to identify the aspect of similarity among the elements in each clusters. We observed the clusters to be formed based on the similarity in the topic being discussed in questions. To name a few, the clusters contained topics corresponding to *MP3 files*, *SQLAlchemy*, *Can’t find something on google*, *How to do something in JavaScript*, and so on.

4 RESULTS

Table 1 shows some examples of the resultant clusters from SOCluster, demonstrating the similarity in terms of the intent, in spite of variation in the textual sentences of the questions. The resultant clusters obtained across the experiments were of different sizes.

Table 1: Example Clusters generated through SOCluster

	Cluster of Questions
1	1413690 : I want to save my countries local time(new zealand) in my database. I am using php but i dont know how to do it. Please guide me out. 1609172 : I want to calculate my countrys((new zealand) local date and time in php but I dont know how to calculate. Please anyone can help me. thanks
2	2089500 : I’m using C++ to get Youtube video links, but now I would like to be able to play a stream in my application. Using C++ wxWidgets, what cross platform options do I have? 2236454 : I would like to stream and play Youtube videos, or FLV files in general, in a C++ application. How can it be done? (I’m using QT as a GUI)
3	2230067 : i want to know whether AST is used in the implementation of search button in eclipse..any idea .help 2234453 : I want to know how AST(Abstract Syntax Tree) is used in the search button of eclipse.Can anyone provide me the link or some info about it .Help

Number before the question indicates the post ID of the question

The spread of the cluster sizes was narrow for both small and big thresholds but it was wide for medium threshold values such as 0.75 & 0.80 . For small thresholds, a big cluster was formed in all cases. For bigger thresholds, maximum clusters had a single element. But medium thresholds such as 0.75 & 0.80 were optimum for the clusters to spread widely. The graphs in Figure 2 summarize the trend of the three evaluation metrics across all experiments. We observed that the results of evaluation are not much affected

by change in the dataset size, but we note that our sample is in the order of 10,000 to 40,000 whereas the number of questions and unanswered questions are in the order of millions on SO.

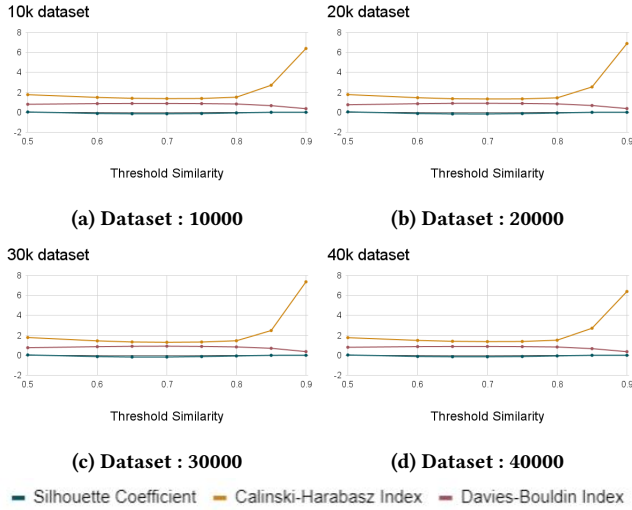


Figure 2: Results of evaluation metrics using SOCluster

However, for a fixed dataset, increase in threshold value beyond 0.7 shows an improvement in the evaluation metric values. The reason behind this might be that threshold similarity of 0.80, 0.85, 0.90 and above impose a very strict similarity condition clustering only near-identical questions together.

DATASET SIZE = 10k									
Threshold Similarity	0.5	0.6	0.65	0.7	0.75	0.8	0.85	0.9	
Structure of the cluster obtained	1 : 25	1 : 459	1 : 1319	1 : 2826	1 : 4842	1 : 7307	1 : 9275	1 : 9919	
	2 : 1	2 : 14	2 : 38	2 : 68	2 : 99	2 : 136	2 : 130	2 : 27	
	9973 : 1	3 : 2	3 : 4	3 : 10	3 : 13	3 : 28	3 : 26	3 : 2	
		6 : 1	4 : 2	4 : 2	4 : 6	4 : 7	4 : 11	4 : 2	
		7 : 1	8 : 1	5 : 1	5 : 1	5 : 3	5 : 1	6 : 1	
		9494 : 1	8577 : 1	12 : 1	6 : 1	6 : 1	8 : 2	7 : 1	
				6983 : 1	7 : 1	7 : 3	9 : 1		
					11 : 2	11 : 1	143 : 1		
					4857 : 1	19 : 1	170 : 1		
						2237 : 1			
Threshold Similarity	0.5	0.6	0.65	0.7	0.75	0.8	0.85	0.9	
Silhouette Coefficient	0.029740976	-0.11862005	-0.1485919	-0.15045904	-0.123852454	-0.06230934	-0.001745860	0.0019306712	
Calinski-Harabasz Index	1.764718781	1.489440095	1.400971023	1.370043814	1.38287018	1.512206407	2.711144878	6.39279339	
Davies-Bouldin Index	0.8016297127	0.8681086244	0.8799939647	0.8809646438	0.8673776209	0.8297854441	0.663908863	0.3636455176	

Figure 3: Distribution of Clusters and evaluation metrics for 10k dataset using SOCluster - Format X : Y where X = cluster size and Y = cluster count

Table in Figure 3 shows the result table for dataset of 10k questions. The detailed results of clustering of different dataset sizes with varying threshold values can be found at this link⁶.

A closer look over the the spread of the clusters over different sizes tells how "good" a cluster is for our intended goal of answering unanswered questions. More evenly spread cluster indicates a better chance of the unanswered questions falling in group with an answered question. If there are too many specific and different

questions which lead to small size clusters, or large clusters, the advantages of clustering would be compromised. Clusters with a more even spread over different sizes indicate the potential of the dataset in handling some of the unanswered questions in the dataset by itself. Changing the dataset to include questions which are more related to each other (belonging to same domain or language) is a solution for this. In our case, this trend can be seen for very high threshold similarity values. Also, large clusters can be troublesome too, if they group most of the questions together, clustering would not be able to serve its purpose. So, the aim should be to achieve clusters with a more even spread over different sizes. It means the dataset has potential for handling some of its unanswered questions by itself. Thus, developers can also use SOCluster's results to evaluate their dataset and then, can work on those datasets towards automatic answering.

5 DISCUSSION AND LIMITATIONS

We aimed at clustering questions on SO platform with the ultimate goal of answering millions of unanswered questions using answered questions. We applied the Sentence-BERT based clustering on sample datasets, with number of questions ranging from 10,000 to 40,000. We observed clusters of varying sizes during this process, which helped us in identifying relevant answers for unanswered questions. However, we also witnessed singleton clusters i.e., clusters with only one entity. On closer examination of the singleton cluster ratio (SCRs), which is the ratio of singleton clusters to total output clusters for different dataset sizes, we observed the SCR to be higher for datasets with smaller sizes. When the dataset size is too small (<20 questions), the SCR is too high (~1, which means mostly singleton cluster) as the odds of two or more questions being similar is low for lesser number of questions. With the increase in dataset size, the SCR remained close to 1 (decreasing very slowly) in our experiment. But, as we further increase the size of the dataset (>40,000 questions), we envision that the SCR might shrink, given higher chances of similarity between two or more questions. However, using a serial BFS traversal algorithm becomes a bottleneck with respect to time.

For the same dataset, when we increased the threshold, the spread of clusters widened, and then it reached to a maxima, and then again narrowed down for higher values of the threshold. The results also indicate that the chances of two questions to fall in the same cluster decreases as the value of threshold increases. Ideal value of the threshold according to our experiment lies in between 0.8 to 0.9. Threshold values greater than 0.9 will be too strict for clustering the questions. Also, lower values of thresholds (<0.5) resulted in clustering of unsimilar questions.

An inherent limitation of the tool is not to consider questions having code snippets (multiple functions/methods) or images in order to avoid the overhead of processing images or code snippets. Wu et al.[23] observed that 75% of the answers on SO have at least one source code snippet attached. We believe that the tool can be extended and integrated with code vectorization, code summarization or image processing techniques. Also, adding a UI or plugin for SOCluster can make it more user-friendly and easy-to-use. Finally, the algorithm and metrics we used for clustering could be further improved, along with validation from user studies.

⁶https://docs.google.com/spreadsheets/d/1W3BAtn6g3-76W0ZQXA7SWiXJ_MvxdF1PjLBSwkW6uE/edit?usp=sharing

This preliminary tool being a first step towards intent-based clustering of SO questions, the questions with code snippets (multiple functions/methods) or images could be considered in the future, owing to the current processing overhead limitations. Similarly, we see a definite scope to leverage different techniques that could improve clustering algorithms and use optimized embeddings for intent recognition based on SE specific domain knowledge. Also, adding a UI or plugin for SOCluster could make it more user-friendly and easy-to-use.

6 RELATED WORK

Clustering of large datasets from Q&A sites has attracted the attention of many researchers. Existing literature has a lot of applications in Question & Answering [11], Dialog System Applications [6] and other domains. Many earlier works have contributed to this through various algorithms and models.

Chen et al. [3] have used a semi-supervised learning technique called *co-training* approach on *Yahoo! Answers* corpus to understand user intent by classifying them into subjective, objective and social. But using these predefined cluster-labels has its limitation of missing out on some important classes. Nie et al. [14] proposed the Constrained Laplacian Rank Algorithm for Graph-Based Clustering. It takes the data graph as input and allows it to be adjusted as part of the clustering procedure so that the quality of the resulting clustering is not affected by low quality input data graph. This CLR-algorithm has been applied and tested against SO dataset by Huang et al. [8], where they have used term frequency based representation of posts. However, their term-frequency based representation does not capture the essence of the *Intent* of the questions.

Yelmen & Duru [24] used K-Means++, K-Medoids & Gaussian Mixture to perform clustering of SO posts using doc2vec, and recent study shows that BERT models have outperformed doc2vec on cybersecurity-related NLP tasks [13]. Our approach uses SentenceBERT vectorizer which has been observed to perform better than BERT [17]. Unlike the earlier machine learning or LDA-based approaches [2, 22] for intent-based classification of SO posts, SOCluster uses graph-based approach for intent-based clustering.

7 CONCLUSION AND FUTURE WORK

In this paper, we proposed SOCluster, an approach and a tool that clusters SO Q&A dataset using graph-based clustering approach and tries to recommend answers to unanswered SO questions. For the demonstration, the tool inputs SO posts with varying sizes and clusters for 8 different threshold values of similarity index and outputs *clusters* containing both answered and unanswered questions. We demonstrated through the SOCluster tool that a graph-based approach for intent-based clustering could have a potential to answer unanswered questions on the SO platform.

Our future goal is to improve the evaluation for optimum cluster-size distribution. We plan to improve the vectorization of the questions by training the model and also plan to include the intent of image or code snippets and cluster the posts which include them. We also plan to consider forming cluster on larger dataset and to implement a parallel version of the BFS algorithm to reduce the run-time of the clustering process.

REFERENCES

- [1] Muhammad Asaduzzaman, Ahmed Shah Mashiyat, Chanchal K Roy, and Kevin A Schneider. 2013. Answering questions about unanswered questions of stack overflow. In *2013 10th Working Conference on Mining Software Repositories (MSR)*. IEEE, 97–100.
- [2] Stefanie Beyer, Christian Macho, Massimiliano Di Penta, and Martin Pinzger. 2018. Automatically classifying posts into question categories on stack overflow. In *2018 IEEE/ACM 26th International Conference on Program Comprehension (ICPC)*. IEEE, 211–21110.
- [3] Long Chen, Dell Zhang, and Levene Mark. 2012. Understanding user intent in community question answering. In *Proceedings of the 21st international conference on world wide web*. 823–828.
- [4] Qian Chen, Zhu Zhuo, and Wen Wang. 2019. Bert for joint intent classification and slot filling. *arXiv preprint arXiv:1902.10909* (2019).
- [5] Themistoklis Diamantopoulos, Maria Ioanna Sifaki, and Andreas Symeonidis. 2019. Towards mining answer edits to extract evolution patterns in Stack Overflow. In *2019 IEEE/ACM 16th International Conference on Mining Software Repositories (MSR)*. IEEE, 215–219.
- [6] Iryna Haponchyk, Antonio Uva, Seunghak Yu, Olga Uryupina, and Alessandro Moschitti. 2018. Supervised clustering of questions into intents for dialog system applications. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. 2310–2321.
- [7] Homa B Hashemi, Amir Asiaee, and Reiner Kraft. 2016. Query intent detection using convolutional neural networks. In *International Conference on Web Search and Data Mining, Workshop on Query Understanding*.
- [8] Chaoran Huang, Lina Yao, Xianzhi Wang, Boualem Benatallah, and Quan Z Sheng. 2017. Expert as a service: Software expert recommendation via knowledge domain embeddings in stack overflow. In *2017 IEEE International Conference on Web Services (ICWS)*. IEEE, 317–324.
- [9] Matthew Huggins, Sharifa Alghowinem, Sooyeon Jeong, Pedro Colon-Hernandez, Cynthia Breazeal, and Hae Won Park. 2021. Practical Guidelines for Intent Recognition: BERT with Minimal Training Data Evaluated in Real-World HRI Application. In *Proceedings of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*. 341–350.
- [10] Ahmed Adeeb Jalal and Basheer Husham Ali. 2021. Text documents clustering using data mining techniques. *International Journal of Electrical & Computer Engineering (2088-8708)* 11, 1 (2021).
- [11] Blooma Mohan John, Alton YK Chua, Dion Hoe Lian Goh, and Nilmini Wickramasinghe. 2016. Graph-based cluster analysis to identify similar questions: A design science approach. *Journal of the Association for Information Systems* 17, 9 (2016), 2.
- [12] Ashish Kathuria, Bernard J Jansen, Carolyn Hafernik, and Amanda Spink. 2010. Classifying the user intent of web queries using k-means clustering. *Internet Research* (2010).
- [13] Otgonpurev Mendsaikhan, Hirokazu Hasegawa, Yukiko Yamaguchi, Hajime Shimada, and Enkhbold Bataa. 2020. Identification of cybersecurity specific content using different language models. *Journal of Information Processing* 28 (2020), 623–632.
- [14] Feiping Nie, Xiaoqian Wang, Michael I Jordan, and Heng Huang. 2016. The constrained laplacian rank algorithm for graph-based clustering. In *AAAI*. Citeseer, 1969–1976.
- [15] Mohammad Masudur Rahman and Chanchal K Roy. 2015. An insight into the unresolved questions at stack overflow. In *2015 IEEE/ACM 12th Working Conference on Mining Software Repositories*. IEEE, 426–429.
- [16] Rajesh N Rao and Manojit Chakraborty. 2021. Vec2GC-A Graph Based Clustering Method for Text Representations. *arXiv preprint arXiv:2104.09439* (2021).
- [17] Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084* (2019).
- [18] Muhammad Roman, Abdul Shahid, Shafiqullah Khan, Anis Koubaa, and Lisu Yu. 2021. Citation intent classification using word embedding. *IEEE Access* 9 (2021), 9982–9995.
- [19] Gilad Tsur, Yuval Pinter, Idan Szpektor, and David Carmel. 2016. Identifying web queries with question intent. In *Proceedings of the 25th International Conference on World Wide Web*. 783–793.
- [20] Ramazan Ünli and Petros Xanthopoulos. 2019. Estimating the number of clusters in a dataset via consensus clustering. *Expert Systems with Applications* 125 (2019), 33–39.
- [21] Ashok Veilumuthu and Parthasarathy Ramachandran. 2009. Intent based clustering of search engine query log. In *2009 IEEE International Conference on Automation Science and Engineering*. IEEE, 647–652.
- [22] Akhila Sri Manasa Venigalla, Chaitanya S Lakkundi, and Sridhar Chimalakonda. 2019. SOTagger-Towards Classifying Stack Overflow Posts through Contextual Tagging (S). In *SEKE*. 493–639.
- [23] Yuhao Wu, Shaowei Wang, Cor-Paul Bezemer, and Katsuro Inoue. 2019. How do developers utilize source code from stack overflow? *Empirical Software Engineering* 24, 2 (2019), 637–673.
- [24] Eray YELMEN and Necihan DURU. 2020. Doc2vec Approach for Text Document Clustering. *ICATCES 2020 Proceeding Book* (2020), 51.